

Dr. Ankit Barik

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Experience

Dept. of Earth & Planetary Sciences, Johns Hopkins University <i>Assistant Research Scientist</i>	Baltimore, USA Nov 2022 – Present
Dept. of Earth & Planetary Sciences, Johns Hopkins University <i>Postdoctoral researcher</i>	Baltimore, USA Nov 2017 – Nov 2022
Max Planck Institute for Solar System Research <i>Postdoctoral researcher</i>	Göttingen, Germany May 2017 – Oct 2017

Education

International Max Planck Research School for Solar System Science <i>PhD, Magna cum laude</i>	Göttingen, Germany 2013 – 2017
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- Thesis title: Inertial modes, turbulence and magnetic effects on a differentially rotating spherical shell
- Thesis supervisors: Dr. Johannes Wicht, Prof. Dr. Ulrich R. Christensen, Prof. Dr. Andreas Tilgner
- Defence date: 08 May, 2017

Indian Institute of Technology, Kharagpur <i>Bachelor's + Master's</i>	Kharagpur, India 2008 – 2013
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- Major: Exploration Geophysics, Minor: Physics
- Thesis title: Effect of gravity environment on dynamo action in rotating spherical shells
- Thesis supervisors: Dr. Johannes Wicht, Prof. W.K. Mohanty

Summer/Winter Schools

- 12th International School/Symposium for Space Simulations (ISSS-12), Prague, Czech Republic, July 2 – 6, 2015
- 'Turbulence, magnetic fields and self organization in laboratory and astrophysical plasmas', Les Houches, France, March 23 – April 03, 2015

Grants/Awards

Visiting positions

2024	Invited professorship at École Centrale Méditerranée/IRPHE, Marseille, France. Total funding received : \$10,000
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Research Grants

2024	Collaborator, NSF Astronomy and Astrophysics Research Grants (AAG)
2020	(Co-wrote) NASA grant proposal for the Cassini Data Analysis Program : \$488,710

Teaching

2023	Cloos Memorial Lecturer, Department of Earth & Planetary Sciences, Johns Hopkins University : \$7,535.
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Computing time

2025	PI, NSF ACCESS computing time: 2.6 million cpu-hours
2024	PI, NSF ACCESS computing time: 1.5 million cpu-hours
Jul 2015 – Jul 2020	Computational time at the North-German Supercomputing Alliance : \$600,000

Pending

2025	PI, NASA Solar System Science (S3)
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Applied but unsuccessful.....

2025	Co-I, NASA Interdisciplinary Consortia for Astrobiology Research (ICAR)
2024	Co-I, NASA Cassini Data Analysis Program (CDAP)
2022	Subaward, NSF Astronomy and Astrophysics Research Grants (AAG)

Other.....

2021	Postdoctoral science teaching fellowship for course "Stellar & Planetary Waves"
2013	Among top 1% selected for a program for fully sponsored PhD in computational sciences by Shell. (declined)
2013	Best Master's Thesis, Department of Geology & Geophysics, IIT Kharagpur
2012	Ranked 2 nd in India in Schlumberger's coding competition for a seismic inversion plug-in for their software 'Petrel'

Mission involvement

Part of science team of the InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport) Mars mission.

Code development

- **MagIC** : 3D pseudo-spectral magnetohydrodynamics (MHD) code to study planetary and stellar interiors. Community code used in over 100 publications. (<https://github.com/magic-sph/magic>)
- **Kore** : 3D Spectral MHD eigenvalue code. (<https://github.com/repepo/kore>)
- **Kaiju** : 3D finite volume MHD code to study magnetospheres (<https://kaiju-docs.readthedocs.io/>)
- **planetMagFields** : Teaching/research tool to visualize magnetic fields of planets in our solar system. (<https://ankitbarik.github.io/planetMagFields>).
- **inermodz** : Python package to compute and plot analytical inertial eigenmodes of a sphere (<https://github.com/AnkitBarik/inermodz>).

Publications

Articles.....

- [1] J. R. Fuentes, **A. Barik**, and J. Fuller. Excitation of Inertial Modes in 3D Simulations of Rotating Convection in Planets and Stars. *The Astrophysical Journal*, 998(1):131, February 2026.
- [2] C. Yan, **A. Barik**, S. Stanley, A. Mittelholz, A.-C. Plesa, and C.-L. Johnson. Mars' hemispheric magnetic field from a full-sphere dynamo. *Geophysical Research Letters*, 52(3):e2024GL113926, 2025.
- [3] **A. Barik**, S. A. Triana, M. Hoff, and J. Wicht. Transition to turbulence in the wide-gap spherical Couette system. *Journal of Fluid Mechanics*, 1001:A1, December 2024.
- [4] **A. Barik** and R. Angappan. planetmagfields: A python package for analyzing and plotting planetary magnetic field data. *Journal of Open Source Software*, 9(97):6677, 2024.
- [5] F. Seuren, S. A. Triana, J. Rekier, **A. Barik**, and T. Van Hoolst. Effects of the Librationally Induced Flow in Mercury's Fluid Core with an Outer Stably Stratified Layer. *The Planetary Science Journal*, 4(9):161, September 2023.
- [6] C. Yan, **A. Barik**, S. Stanley, J. Leung, A. Mittelholz, C. L. Johnson, A.-C. Plesa, and A. Rivoldini. An ancient martian dynamo driven by hemispheric heating: effect of thermal boundary conditions. *Planetary Science Journal*, 2023.
- [7] T. Gastine, **A. Barik**., rraynaud, t schwaiger, B. Putigny, thtassin, J. Wicht, L. Duarte, and B. Dintrans. magic-sph/magic: release magic 6.2, December 2022.
- [8] **A. Barik**, S. A. Triana, M. Calkins, Stanley S., and J. Aurnou. Onset of convection in rotating spherical shells: Variations with radius ratio. *Earth and Space Science*, 2022.

- [9] K. M. Moore, **A. Barik**, S. Stanley, D. J. Stevenson, N. Nettelmann, R. Helled, T. Guillot, B. Militzer, and S. Bolton. Jupiter's dynamo magnetic field: The role of stable stratification and a dilute core. *Journal of Geophysical Research: Planets*, 2022.
- [10] S. A. Triana, G. Guerrero, **A. Barik**, and J. Rekier. Identification of inertial modes in the solar convection zone. *The Astrophysical Journal Letters*, jul 2022.
- [11] B. J. Anderson, R. Angappan, **A. Barik**, S. K. Vines, S. Stanley, P. N. Bernasconi, H. Korth, and R. J. Barnes. Iridium Communications Satellite Constellation Data for Study of Earth's Magnetic Field. *Geochemistry, Geophysics, Geosystems*, August 2021, **Highlighted by the Nature magazine** (<https://www.nature.com/articles/d41586-021-01860-9>).
- [12] V. Perera, C. Mead, K. J. van der Hoeven Kraft, S. Stanley, R. Angappan, S. MacKenzie, **A. Barik**, and S. Buxner. Considering intergroup emotions to improve diversity and inclusion in the geosciences. *Journal of Geoscience Education*, July 2021.
- [13] **A. Barik**, S. A. Triana, M. Hoff, and J. Wicht. Triadic resonances in the wide-gap spherical couette system. *Journal of Fluid Mechanics*, 2018.

Book Chapters.....

- [14] M. Le Bars, **A. Barik**, F. Burmann, D. P. Lathrop, J. Noir, N. Schaeffer, and S. A. Triana. Fluid Dynamics Experiments for Planetary Interiors. *Surveys in Geophysics*, December 2021.

Submitted/Under review.....

- [15] L. Ojha, **A. Barik**, and J. Buffo. On the feasibility of a direct fluid exchange between the deep subsurface ocean and the shallow subsurface environment of Europa.
- [16] J. Rekier, S. A. Triana, **A. Barik**, D. Abdullah, and W. Kang. Kilometer-high mountains at the bottom of earth's mantle revealed by nutation.

Soon to be submitted.....

- [17] **A. Barik**, S. Stanley, B. Tian, S. Tikoo, and B. Weiss. An ancient lunar dynamo driven by mantle precession and convection.
- [18] R. Angappan, **A. Barik**, B. J. Anderson, S. K. Vines, and Stanley S. Fast global wave detection in geomagnetic jerk occurrences with commercial satellites.
- [19] S. A. Triana, J. Rekier, F. Gerick, **A. Barik**, and V. Dehant. Torsional Alfvén modes in the earth's core: A numerical model.

Selected talks and Posters

Invited talks at conferences/workshops.....

2024, Aug 26-30	Effect of the Mantle on Planetary Core-driven Dynamos, <i>Magnetism, Atmospheres, and Life workshop</i> , University of Illinois Urbana-Champaign
2022, Dec 12-16	Comparison of Jupiter's and Saturn's magnetic fields and implications for their interiors, <i>AGU Fall Meeting 2022</i> , Chicago
2022, Jul 14	Effect of libration on a stable layer: an application to Mercury, <i>17th SEDI symposium</i> , ETH Zurich
2021, Jul 27	planetMagFields : A python package for planetary magnetic fields, <i>OpenPlanetary Virtual Lunches</i> , Virtual
2021, Jul 08	Inertial modes from differential rotation: the spherical Couette system, <i>Kavli Summer Program in Astrophysics 2021</i>
2020, Sep	Dynamos driven by convection and precession, <i>17th Symposium of Study of the Earth's Deep Interior (SEDI)</i> , Virtual
2020, Sep 1-4	Triadic resonances in the spherical Couette system, <i>ISSI workshop on Deep Earth</i> , Bern, Switzerland

- 2017, Feb 27-28 Inertial and magneto-Coriolis modes in the spherical Couette flow, *3rd ANR IMAGINE Meeting*, L'Institut de Recherche en Astrophysique et Planétologie (IRAP), Toulouse, France
- 2014, Dec 13 Spherical Couette flow simulations, *Workshop on Geomagnetic Prediction*, hosted by the CIDER project, UC Berkley, Berkley, California, USA

Invited seminars.....

- 2025, March 4 Invited seminar at Southwest Research Institute (SwRI), Boulder, Colorado, USA
- 2025, Feb 25 Center for Environmental and Applied Fluid Mechanics (CEAFM) seminar, Johns Hopkins University, USA
- 2024, Nov 12 Seminar at Institut de Recherche sur les Phénomènes Hors Equilibre, Marseille
- 2024, Nov 4/5 Hosted by Henri-Claude Nataf at ISTerre, Grenoble
- 2024, Oct 21/22 Hosted by Hagay Amit at University of Nantes
- 2024, Sep *Geophysical/Astrophysical Fluid Dynamics Seminar*, University of Colorado Boulder
- 2023, Oct 11 *Planetary Lunch Series*, Massachusetts Institute of Technology
- 2022, Oct 27 *Fluids & MHD seminar*, University of Leeds
- 2022, May 27 *IGPP Seminar*, UC Santa Cruz
- 2021, Nov 11 *EPM Group Seminar*, ETH Zurich
- 2018, Feb 15 *Applied Dynamics Seminar Series*, University of Maryland, College Park, USA

Contributed talks.....

- 2021, Dec 13-17 Onset of convection in rotating spherical shells, *AGU Fall Meeting 2021*
- 2020, Dec 1-17 The ancient lunar dynamo, *AGU Fall Meeting 2020*, Virtual
- 2019, Dec 9-13 Inertial Wave Generation from Boundary Layer Turbulence, *AGU Fall Meeting 2019*, San Francisco, USA
- 2019, May 20-22 A Lunar dynamo driven by mantle precession and convection, *Core of the Moon workshop*, Marseille, France
- 2017, Jun 25-Jul 1 Triadic resonances in the spherical Couette flow, *2nd Conference on Natural Dynamos*, Valtice, Czech Republic
- 2017, Jun 25-Jul 1 Spherical Couette dynamos, *2nd Conference on Natural Dynamos*, Valtice, Czech Republic
- 2015, Jun 22-24 Flow instabilities in the Spherical Couette System, *19th International Couette-Taylor Workshop*, Brandenburg University of Technology, Cottbus, Germany

Contributed posters.....

- 2025, Dec 15-19 Rotating convection with topography: Implications for Core-Mantle and ice-ocean boundaries, *AGU Fall Meeting 2024*, New Orleans, USA
- 2025, Sep 7-12 Topography at the ice-ocean boundary: effects on heat transport, *EPSC-DPS Joint Meeting 2025*, Helsinki, Finland
- 2025, Jan 27-31 Topography at the core-mantle boundary: effects on heat transport, *IPAM workshop on Rotating Turbulence: Interplay and Separability of Bulk and Boundary Dynamics*, UCLA, Los Angeles, USA
- 2024, Dec 12 planetMagFields: A Python package for analyzing and visualizing planetary magnetic field data, *AGU Fall Meeting 2024*, Washington DC, USA
- 2024, Jun 12-13 Interaction of Convection and Tides in Icy Moons, *OPAG Meeting*, Ithaca, NY, USA
- 2023, Nov 19-21 Kore : A spectral anelastic MHD eigenvalue code for rotating fluids in spherical geometries, *76th Annual Meeting of the Division of Fluid Dynamics*, Washington DC, USA
- 2018, Dec 10-14 A Lunar Dynamo Driven by Mantle Precession and Convection, *AGU Fall Meeting 2018*, Washington DC, USA
- 2018, Jul 8-13 Turbulence in spherical Couette flow and the effect of density stratification, *Study of the Earth's Deep Interior (SEDI) 2018*, Edmonton, Canada
- 2016, Dec 12-16 Identification and onset of inertial modes in the wide-gap spherical Couette system, *AGU Fall Meeting 2016*, San Francisco, USA

Teaching

- March 2021 Certificate of completion - Johns Hopkins "Teaching Academy"
- Attending course "Preparation for university teaching"
 - Attending pedagogy seminars/workshops
 - More than six hours of teaching
- 2024 Sep - Nov **Invited professor** at École Centrale Méditerranée / IRPHE, Marseille, France

Graduate courses.....

- 2026 Spring **Guest lecturer**, "Planetary Interiors", Johns Hopkins University
- 2024 Nov **Guest lecturer** "Environmental Fluid Mechanics", Centrale Méditerranée, Marseille, France
- 2024 Oct **Guest lecturer** "Numerical methods: spectral methods", Centrale Méditerranée, Marseille, France
- 2024 Fall **Guest lecturer** "Magnetic Fields in Planetary Systems", Purdue University
- 2023 Fall **Cloos Memorial Lecturer** "Earth and Planetary Fluids", Johns Hopkins University
- 2023 Spring **Guest lecturer**, "Planetary Interiors", Johns Hopkins University
- 2021 Spring **Guest lecturer**, "Planetary Interiors", Johns Hopkins University
- 2021 Spring **Guest lecturer**, "Special topics in dynamo theory", Johns Hopkins University
- 2019 Fall **Guest lecturer**, "Earth and Planetary Fluids I", Johns Hopkins University
- 2019 Spring **Guest lecturer**, "Planetary Interiors", Johns Hopkins University
- 2014 Fall **Teaching assistant**, "Solar System Science: The Central Star", University of Göttingen

Undergraduate courses.....

- 2014 Spring **Teaching assistant**, "Computational Physics", University of Göttingen
- 2014 Spring **Teaching assistant**, "Introduction to Astro-and Geophysics", University of Göttingen

Other.....

- 2015 Nov 4-6 **Tutor**, hands-on workshop on 'MagIC' code "Dynamos in a Nutshell"

Mentoring and supervision

Graduate students.....

- Hachem Dhouib, PhD student at CEA Saclay, for Kavli Summer Program in Astrophysics 2021, June 7th to July 16th, 2021
 - Project: Angular momentum transport by gravito-inertial waves in intermediate-mass stars
- PhD students in the research group: Chi Yan (graduated), Regupathi Angappan (graduated)

Undergraduate students.....

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| Fall 2018 | Nina Amezcua | Exoplanet magnetic fields |
| Fall 2018 | Mackenzie Mills | Ancient martian dynamo |
| Summer 2020 | Brian Song | (co-advising) Magnetic data from Iridium Satellites |
| Summer 2021 | Nick Lu | (co-advising) Magnetospheric simulations of the Earth |
| Summer 2021 | Vishnu Srinivasan | (co-advising) Spherical harmonic transforms, use of MagIC simulation code |

Professional services

Proposal review

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| ○ NASA, 2025 | ○ Czech Science Foundation, 2024 |
| ○ Swiss National Supercomputing Centre (CSCS), 2026 | ○ ETH Zurich, Sep 2022 |
| ○ Swiss National Supercomputing Centre (CSCS), 2025 | ○ ETH Zurich, May 2022 |
| | ○ NASA, 2020 |
| | ○ NASA, 2019 |

PhD thesis review and defense committee

Albert Elias López, Institute of Space Sciences (ICE-CSIC)/Universitat Autònoma de Barcelona, Barcelona, Spain

Journal referee	○ Nature ○ Nature Geosciences ○ Comptes Rendus Physique ○ European Journal of Mechanics - B/Fluids ○ Geophysical & Astrophysical Fluid Dynamics	○ Earth and Space Science ○ The Astrophysical Journal ○ Journal of Open Source Software ○ Space Science Reviews
Journal referee (cont.)	○ Astronomy & Astrophysics ○ Planetary Science Journal ○ Geophysical Journal International ○ Earth, Planets and Space ○ Geophysical Research Letters	○ International Journal on Geomatics ○ Journal of Geophysical Research: Planets ○ Physics of Fluids
Member	○ Executive committee, web / social media manager for Geomagnetism, Paleomagnetism and Electromagnetism (GPE) Section of AGU ○ American Geophysical Union (AGU) ○ American Physical Society (APS) ○ AAS Division for Planetary Sciences (DPS)	

Conference organisation.....

2024, Dec 9-13	Convener, session "P023 - Oscillations in Internal Fluid Layers of Planets, Moons, and Stars" at AGU Fall Meeting 2024
2024, Dec 9-13	Convener, session "DI003 - Core-Mantle Interactions: The Dynamic Duo Shaping Our Planet" at AGU Fall Meeting 2024
2023, Dec 12	Convener, session "P23G - Oscillations in Internal Fluid Layers of Planets, Moons, and Stars" at AGU Fall Meeting 2023
2016, Nov 30-Dec 2	17 th MHD Days, 88 participants
2015, Nov 22-24	14 th General meeting of PhDnet, 99 participants
2015, Nov 4-6	MagIC code workshop "Dynamos in a Nutshell", 35 participants

Outreach

2025, Sep 16	"Magnetic fields of the planets around us", Lecturers Without Borders (LeWiBo)
2023, Apr 4	AGU "Ask a Scientist" for Earth Day 2023, Washington DC
2020 - present	Social media manager for DIYnamics (Twitter: @DIYnamicsTeam) - an outreach effort from UCLA for studying/demonstrating geophysical fluid dynamics at home/class
2019, Sep/Oct	Outreach video "The Magnetic Fields of the Solar System" (https://www.youtube.com/watch?v=7S_VqFJep_0) - 40k views
2019, Oct 8	Talk "Everything wrong with The Core" - a talk on the movie
2019, Jul 24	Talk "Planetary magnetic fields: where do they all come from?" at the 2019 QuarkNet workshop
2018	Volunteer at the National Air and Space Museum, Washington DC